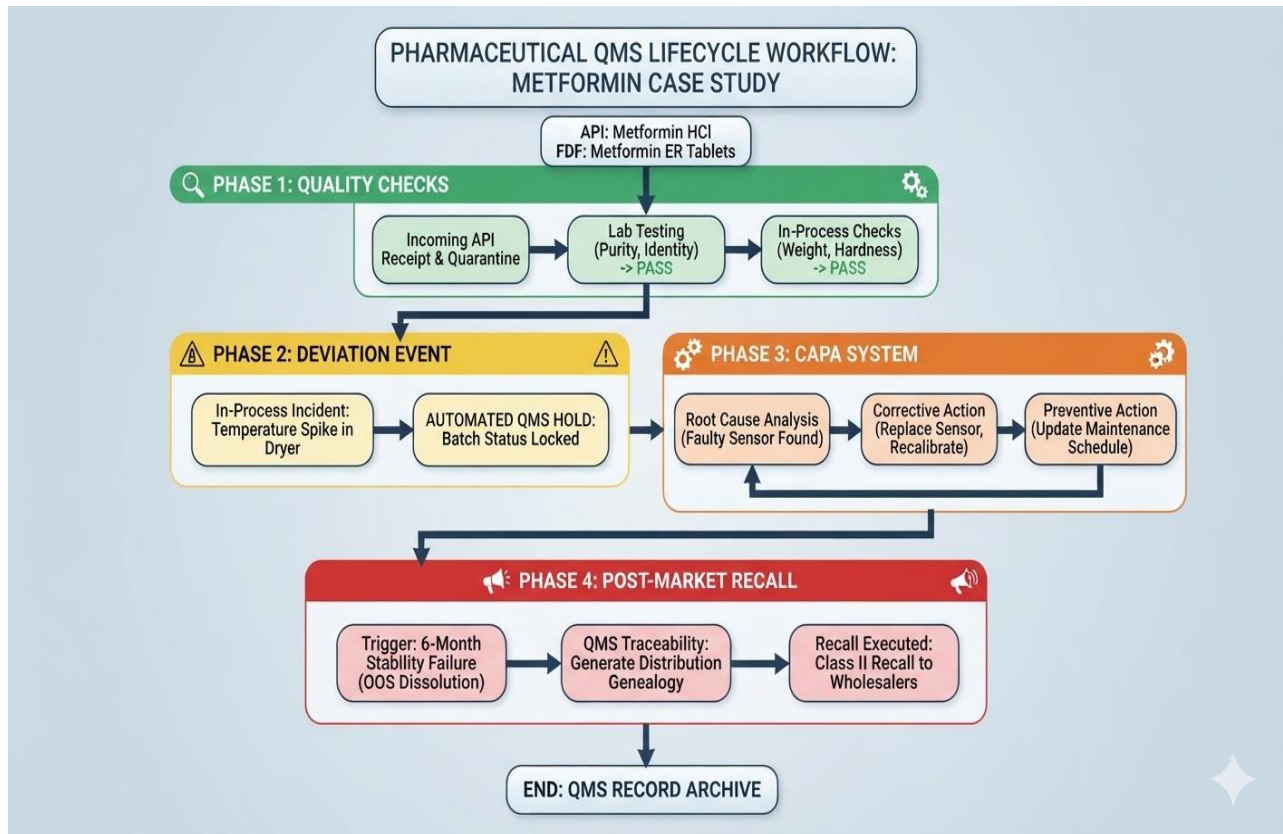


Pharmaceutical QMS Workflow Analysis: Metformin Case Study

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Target Role: Quality Assurance (QA) / Quality Management System (QMS) Associate



1. Product Context & Selection

To demonstrate the practical application of QMS modules in modern pharmaceutical manufacturing, this project maps the lifecycle workflow of a widely prescribed anti-diabetic medication.

- **Active Pharmaceutical Ingredient (API):** Metformin Hydrochloride (Pure crystal powder form).
- **Finished Dosage Form (FDF):** Metformin Extended Release (ER) 500mg Tablets.

2. Core QMS Modules & Lifecycle Workflow

Phase 1: Incoming & In-Process Quality Checks

The QMS serves as the enforcement mechanism ensuring compliance before material advances through the facility.

- **Material Disposition:** Upon receipt of the Metformin Hydrochloride API, the QMS automatically places the lot into a "Quarantine" status. It triggers sampling tasks for laboratory analysts.
- **Testing Protocols:** The QMS logs raw material verification against established specifications (e.g., identity, purity via HPLC, moisture content, and particle size distribution).
- **In-Process Controls (IPC):** During the compression stage of the Metformin ER tablets, the system enforces periodic checks every 30 minutes for critical quality attributes (CQAs) including tablet thickness, weight variation, and friability.
- **System Gatekeeping:** The batch cannot progress to packaging until all IPC parameters are closed and marked as "Pass" within the QMS system.

Phase 2: Deviation Management

When an unexpected event occurs that breaches established manufacturing parameters, the QMS captures and isolates the risk.

- **The Incident:** During the fluid-bed drying phase of the Metformin tablet blend, the system detects a temperature spike exceeding the validated upper limit of 60°C, reaching 68°C for a duration of 14 minutes.
- **Immediate Capture:** The operator initiates an Electronic Deviation Report within the QMS. The system automatically shifts the batch status to "Quality Hold," preventing the ERP system from allowing packaging or distribution.
- **Impact Assessment:** QA utilizes the QMS to launch a cross-functional investigation. The batch is sampled and evaluated to check if the thermal spike caused degradation of the Metformin molecule or altered the extended-release polymer matrix.

Phase 3: Corrective and Preventive Action (CAPA)

The CAPA module ensures that the root cause of the deviation is systematically eliminated to prevent recurrence.

- **Root Cause Analysis (RCA):** Using an integrated 5-Why analysis within the QMS deviation record, the engineering team determines that the primary heating element sensor had suffered a mechanical calibration failure.

- **Corrective Action (CA):** The faulty sensor is immediately tagged, removed, replaced, and verified by an independent calibration engineer.
- **Preventive Action (PA):** The QMS drives systemic change by updating the facility's Preventive Maintenance (PM) schedule to include bi-monthly sensor diagnostic checks instead of bi-annual reviews.
- **Effectiveness Review:** The QMS schedules an automated review task for QA 90 days post-implementation to verify that zero temperature-related drying deviations have recurred.

Phase 4: Market Surveillance & Recall Management

If a defect bypasses internal checks and reaches commercial distribution, the QMS acts as the single source of truth for tracking and recovery.

- **The Trigger:** During routine, ongoing 6-month stability testing of retained samples, a specific batch of Metformin ER tablets demonstrates an out-of-specification (OOS) result for dissolution rate. The tablet dissolves faster than the extended-release window allows.
- **Traceability Mapping:** The QA team utilizes the QMS Genealogy Tree to instantaneously trace every distribution channel, pharmacy chain, and wholesale depot that received shipments from that specific batch.
- **Execution:** A voluntary Class II Recall is initialized. The QMS generates standardized recall notification letters, tracking response rates and physical unit reconciliations as recalled bottles are returned to the warehouse for destruction.